set is Span$\{u, v\}$. Since neither u nor v is a scalar multiple of the other, the solution set is a plane through the origin. □

In Example 1, the general solution of $Ax = 0$ has the following vector form:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{4}{3}x_3 \\ 0 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 4/3 \\ 0 \\ 1 \end{bmatrix} = x_3 v, \text{ where } v = \begin{bmatrix} 4/3 \\ 0 \\ 1 \end{bmatrix}$$

Here x_3 is factored out of the expression for the general solution vector. This shows that every solution of $Ax = 0$ in this case is a scalar multiple of v. The trivial solution is obtained by choosing $x_3 = 0$. Geometrically, the solution set is a line through 0 in $\mathbb{R}^3$.

Important Point

Examples 1 and 2, along with the exercises, illustrate the fact that the solution set of a homogeneous equation $Ax = 0$ can always be expressed explicitly as Span$\{v_1, \ldots, v_p\}$ for suitable vectors $v_1, \ldots, v_p$. If the only solution is the zero vector, then the solution set is Span$\{0\}$. If the equation $Ax = 0$ has only one free variable, the solution set is a line through the origin. A plane through the origin provides a good mental image for the solution set of $Ax = 0$ when there are two or more free variables.

Parametric Vector Form

The original equation (1) for the plane in Example 2 is an $\boxed{\text{implicit}}$ description of the plane. Solving this equation amounts to finding an $\boxed{\text{explicit}}$ description of the plane as the set spanned by u and v. Equation (2) is called a $\boxed{\text{parametric vector equation}}$ of the plane.

Sometimes such an equation is written as $\boxed{x = su + tv \ (s, t \text{ in } \mathbb{R})}$ to emphasize that the parameters vary over all real numbers. In Example 1, the equation $x = x_3 v$ (with x_3 free), or $x = tv$ (with t in $\mathbb{R}$), is a parametric vector equation of a line. Whenever a solution set is described explicitly